

Figure 1. D1 sub-score → grade reference matrix (DO/ORP-only main link, final)

criterion

A (≥4.5)

B (3.5–4.5)

C (2.5–3.5)

D (1.5–2.5)

F (<1.5)

Q_spike
(spike)

spike rate ≤ 2%

very rare

rare

occasional

frequent very frequent (>20%)

Q_step
(step)

KS statistic

near 0

low

moderate

elevated

high (sustained)

Q_drift
(drift)

PLS residual z

|z| < 1.5

1.5–2.0

2.0–2.5

2.5–3.0

> 3.0 sustained

Q_freeze
(freeze)

RLE / unique ratio

no freeze

<5 min

5–15 min

15–60 min

>60 min

Q_regime
(regime)

W1 normalised

< 1.0

1.0–2.0

2.0–3.0

3.0–4.0

> 4.0